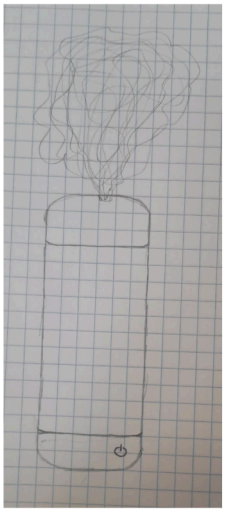
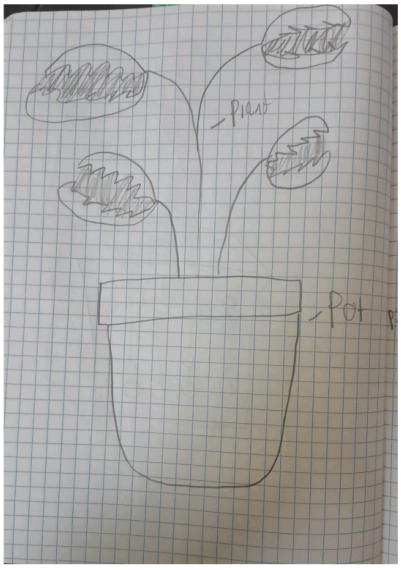
Element D: Design Concept Generation, Analysis, and Selection

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students' ability to rest.

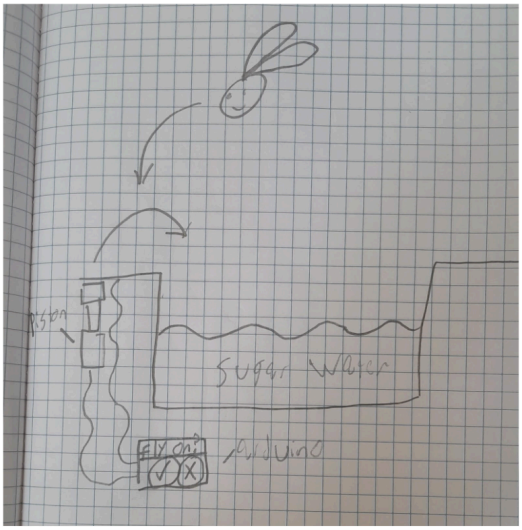
Our team aimed to develop an effective, ethical, and efficient solution for capturing insects. We began by brainstorming potential solutions, documented in our notebooks, considering various mechanisms for attracting and containing insects. This initial phase encouraged a high volume of ideas, without immediate judgment, to help with our creativity. Following the initial brainstorming, we narrowed our ideas by evaluating them against key design requirements and ethical considerations. We considered factors such as safety, ease of use, potential environmental impact, and the ability to target insects. This process led us to focus on several promising concepts that we then formalized as distinct proposed solutions: a diffuser with peppermint oil, a venus fly trap, a bowl of sugar water with platforms, a sticky tube, and an automatic fly catcher.

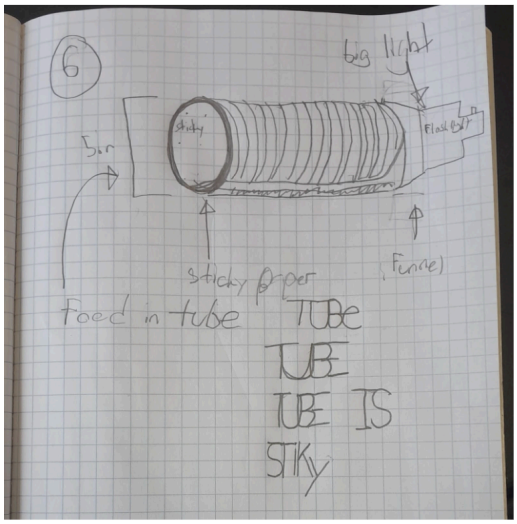


Idea number one : Diffuser with peppermint oil



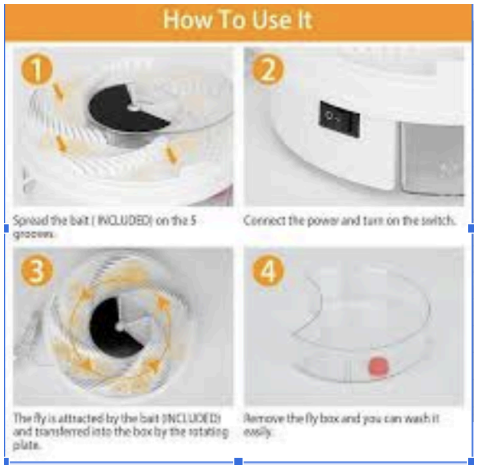
Idea number two: Venus flytrap





Idea number three: bowl of sugar water

Idea number four: Sticky tube



Idea number five: Automatic fly catcher

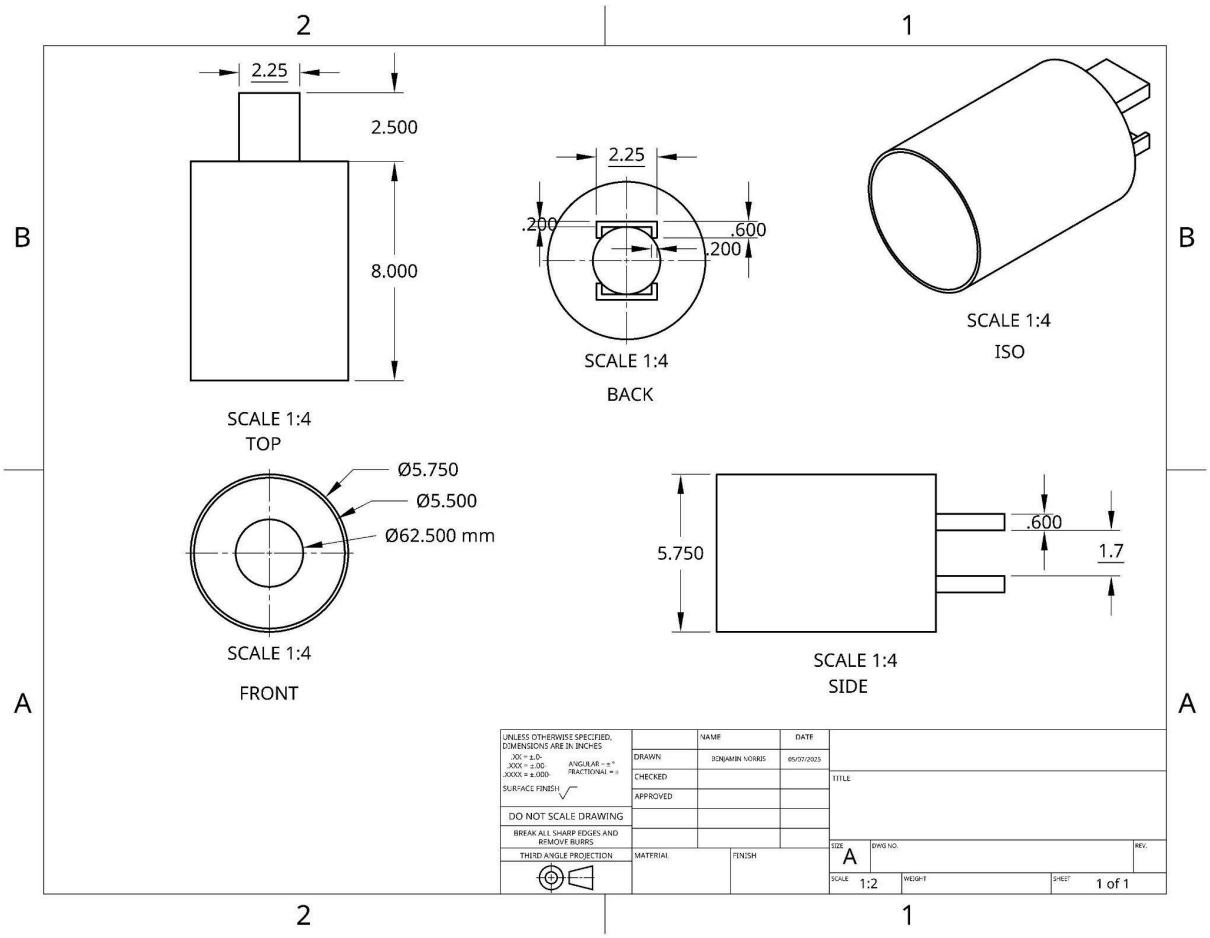
To systematically compare these solutions, we developed a weighted decision matrix. We identified critical design criteria, including effectiveness at capturing insects, safety and efficient use, assembly time, attracting insects without using materials that cause harm to humans, animals, or the environment, and aesthetics. Each prioritized design requirement was assigned a

weighting based on its importance to the overall success of the solution. We then evaluated each proposed solution against these criteria, assigning a score from -2 to +2 (where -2 is significantly worse than the current solution, 0 is the same as the current solution, and +2 is better). The decision matrix allowed us to compare the strengths and weaknesses of each concept.

	Proposed Solutions					
Design Requirements	**Current Solution:** Nothing	**Idea #1:** Diffuser	**Idea #2:** Venus Fly trap	**Idea #3:** Sugar Water trap	**Idea #4:** Swirly portal	**Idea #5:** Sticky tube
Small & efficient	0	+0.5	+0.5	-1	+0.5	+1
Assembly time is 10 minutes	0	+1	+1	-1	-2	+1
Needs to be ethical	0	-1	+1	+1	+1	+1
Attracts bugs to the trap or repels bugs from the area (x2)	0	+1 * 2 = 2	+1 * 2 = 2	+1* 2 = 2	+1* 2 = 2	+1* 2 = 2
Aesthetic	0	0	0	-1	+1	+0
Needs to be long term solution	0	+1	+1	+0.5	+1	+0.5
Feasibility (x2)	0	+1* 2 = 2	+1* 2 = 2	+1* 2 = 2	-2 * 2 = -4	+1* 2 = 2
Total Score	**0**	**5.5**	**6.5**	**2.5**	**-0.5**	**7.5**

The decision matrix

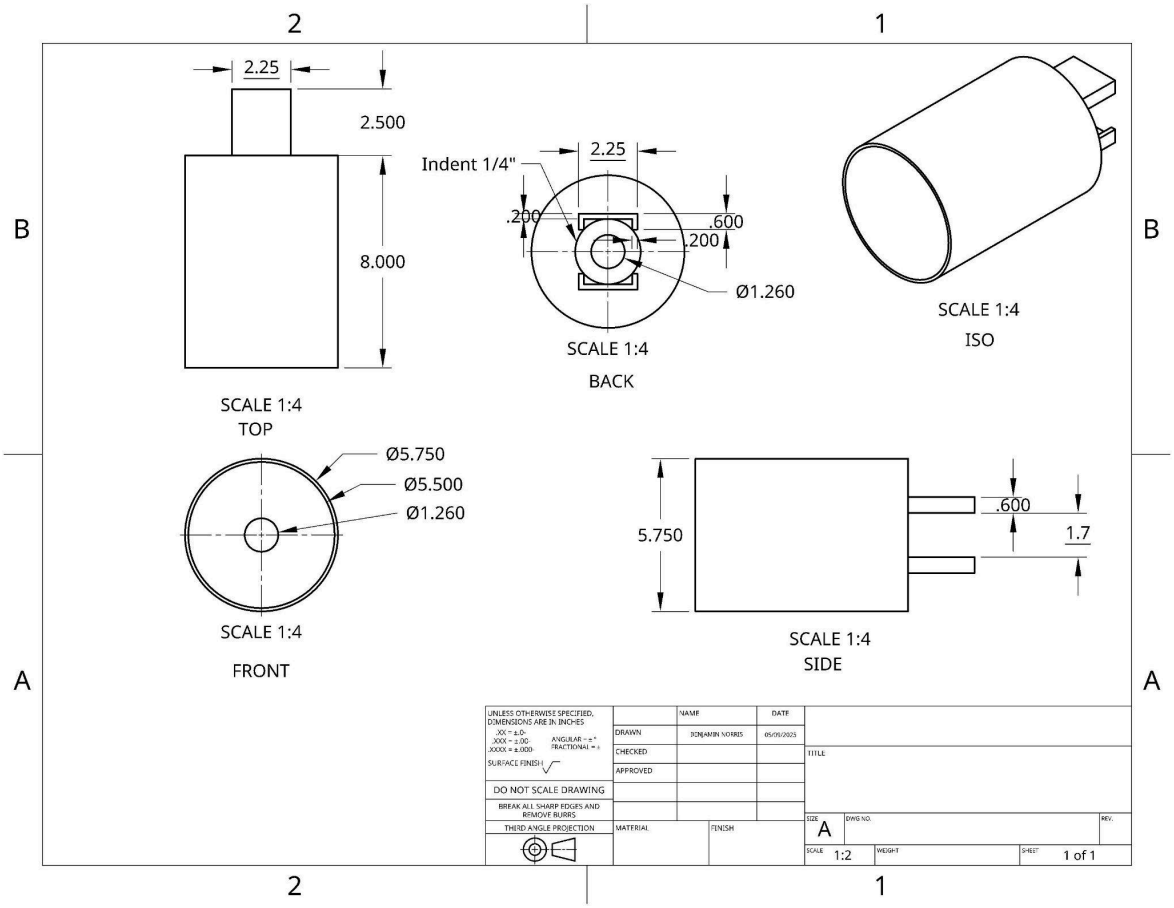
After calculating the weighted total score for each solution, the "Sticky Tube" emerged as the most promising option with a total score of +7.5.



First Prototype

Our second prototype builds upon the original sticky tube concept but utilizes a custom-designed and 3D-printed tube instead of a paper towel tube. This tube, printed with PLA, polylactic acid filament, maintains the same internal dimensions to accommodate a roll of sticky paper and has a total length of 203.2 mm. Similar to the first design, we inserted a small flashlight approximately 63.5 mm into one end. However, we have incorporated key modifications to control the light. We have added a small indent at the flashlight insertion point within the 3D-printed design. This indent helps limit the amount of light escaping the tube. Additionally, an integrated and recycled lens, sourced from a broken projector, is at the opening of the indent in the 3D-printed tube. This lens serves to disperse the flashlight's beam more

evenly inside the tube, rather than allowing it to shine outwards. This modification, enabled by the customizability of 3D printing, aims to improve the trap's efficiency by directing more light onto the sticky surface.



Second Prototype